



A Systematic Comparison of Large and Small Language Models in Standalone, Hybrid & Multi-Agent Architectures

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1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures** in **three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Routing reaches **EATS 0.893** on HellaSwag and **0.854** on TruthfulQA; Pure Swarm leads GSM8K at **EATS 0.841**; Speculative achieves **EATS 0.908** on ARC. Novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.

2 Background / Literature Review

LLMs deliver state-of-the-art reasoning but impose high latency, cost, and energy. **SLMs** are cheaper and data-sovereign, yet trail on broad reasoning tasks.

A systematic literature review of **54 collaborative SLM–LLM studies** indicates that, while hybrid and multi-agent designs are becoming more common, **cost and energy are still rarely treated as primary evaluation outcomes** alongside accuracy.

RQ1 Performance. How do collaborative SLM/LLM systems compare with standalone LLMs on reasoning & classification?

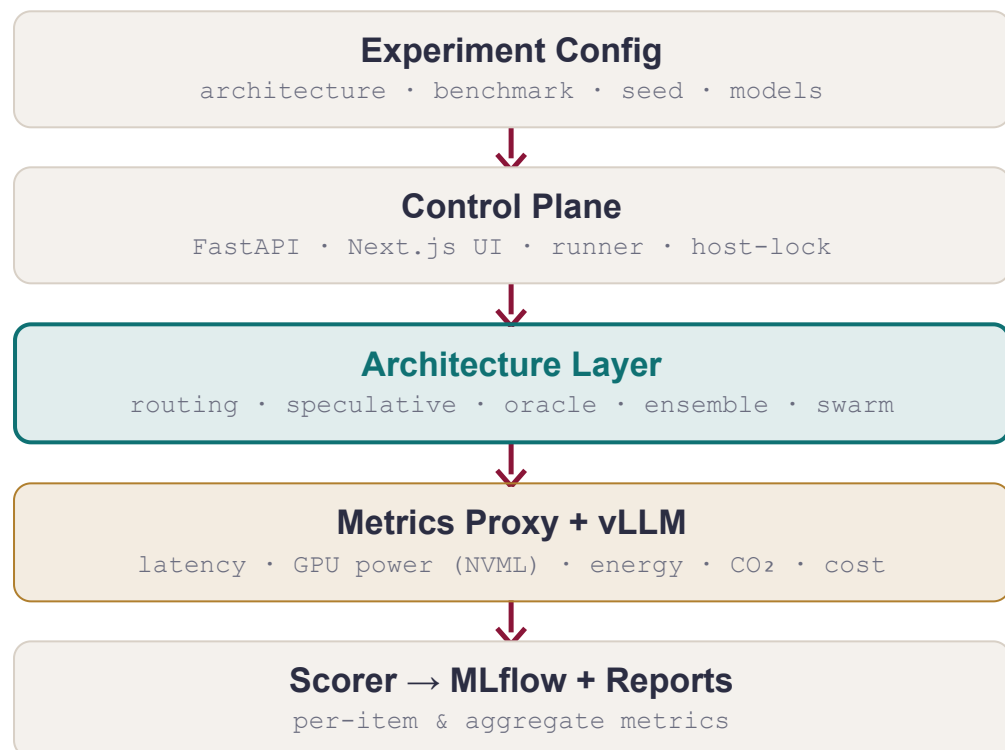
RQ2 Efficiency. How do hybrid & multi-agent designs affect cost, latency, and energy?

RQ3 Unified metric. Does EATS give a stable, architecture-agnostic ranking of the quality–efficiency trade-off?

RQ4 Empirical grounding. Do measured comparisons corroborate the trends from our 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same *Query* and returns the same *Response* carrying **accuracy, latency, tokens, cost, energy & CO₂** per query. The architecture layer is the *only* interchangeable component.



TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	gemma4, qwen3.5, llama3.2 (3–4B)
Mid LLM	NVIDIA RTX PRO6000	96 GB	gpt-oss, gemma4, qwen3.5 (20–35B dense + MoE)
Heavy LLM	NVIDIA H200	141 GB	gpt-oss, llama3.3 (70–120B)

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\text{EATS} = \frac{\text{Accuracy}}{\text{Accuracy} + \lambda \cdot P_{\text{efficiency}} + \beta \cdot P_{\text{accuracy}}}$$

$$P_{\text{efficiency}} = 0.65 \cdot \text{Cost} + 0.20 \cdot \text{Latency} + 0.15 \cdot \text{Energy}$$

$$P_{\text{accuracy}} = 1 - \text{Accuracy}$$

$$\beta = 0.60$$

$$\lambda = 0.40$$

all terms normalised per standalone-LLM baseline · $\beta = 0.60$, $\lambda = 0.40$ calibrated via grid search on MMLU validation split

5 Energy, Carbon & Cost

Per-request measurements via GPU power sampling (NVIDIA NVML) at the inference host.

$$\text{Cost} = \text{GPU rate} \cdot \text{time}$$

$$E = P_{\text{GPU}} \cdot \Delta t$$

$$\text{Latency} = \Delta t \text{ per request}$$

$$\text{CO}_2 = k \cdot E$$

$$k = 380 \text{ gCO}_2/\text{kWh (EU avg)} \cdot E \text{ in J} \div 3.6 \times 10^6$$

6 Benchmarks & Protocol

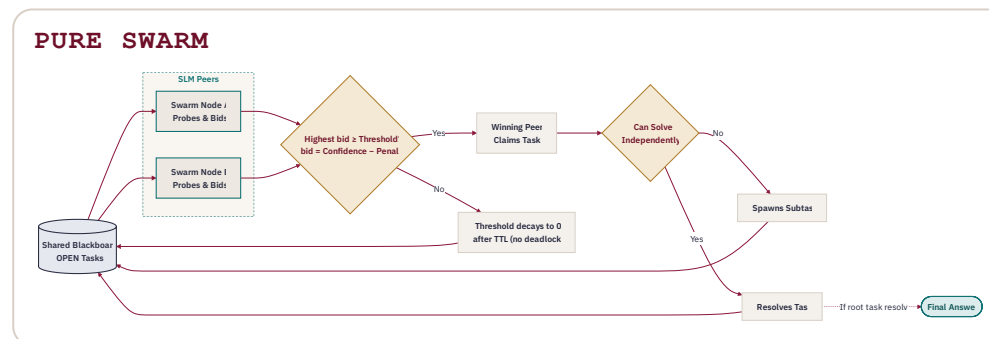
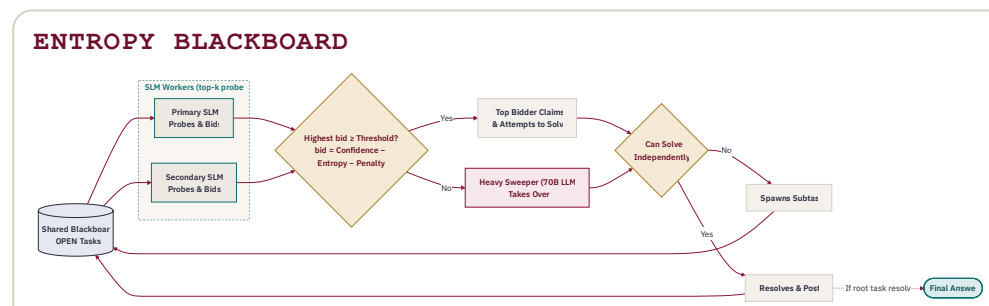
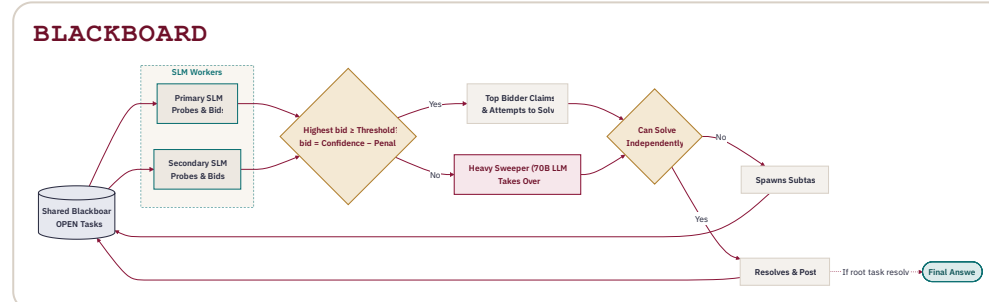
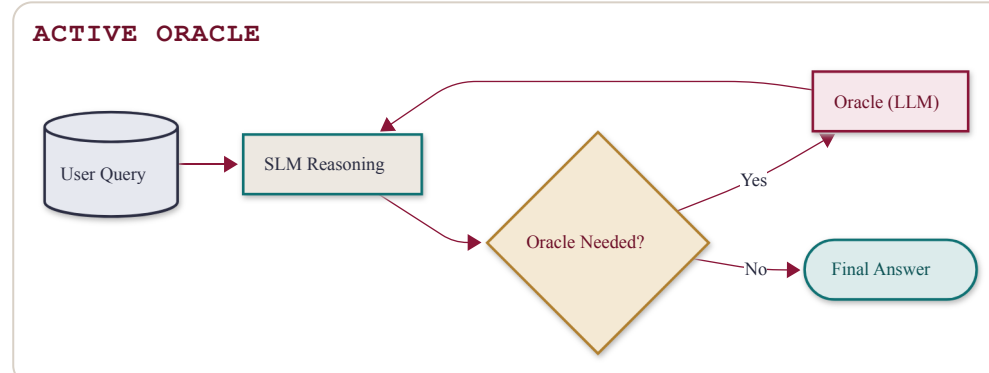
Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking.

BENCHMARK	WHAT IT MEASURES
MMLU	Broad knowledge — 57 academic disciplines
ARC-C	Scientific reasoning — multi-step physical-world science
GSM8K	Mathematical reasoning — grade-school math, free-form
HellaSwag	Commonsense inference — sentence completion
TruthfulQA	Adversarial truthfulness — calibration vs. persuasive falsehoods

7 Architecture Taxonomy

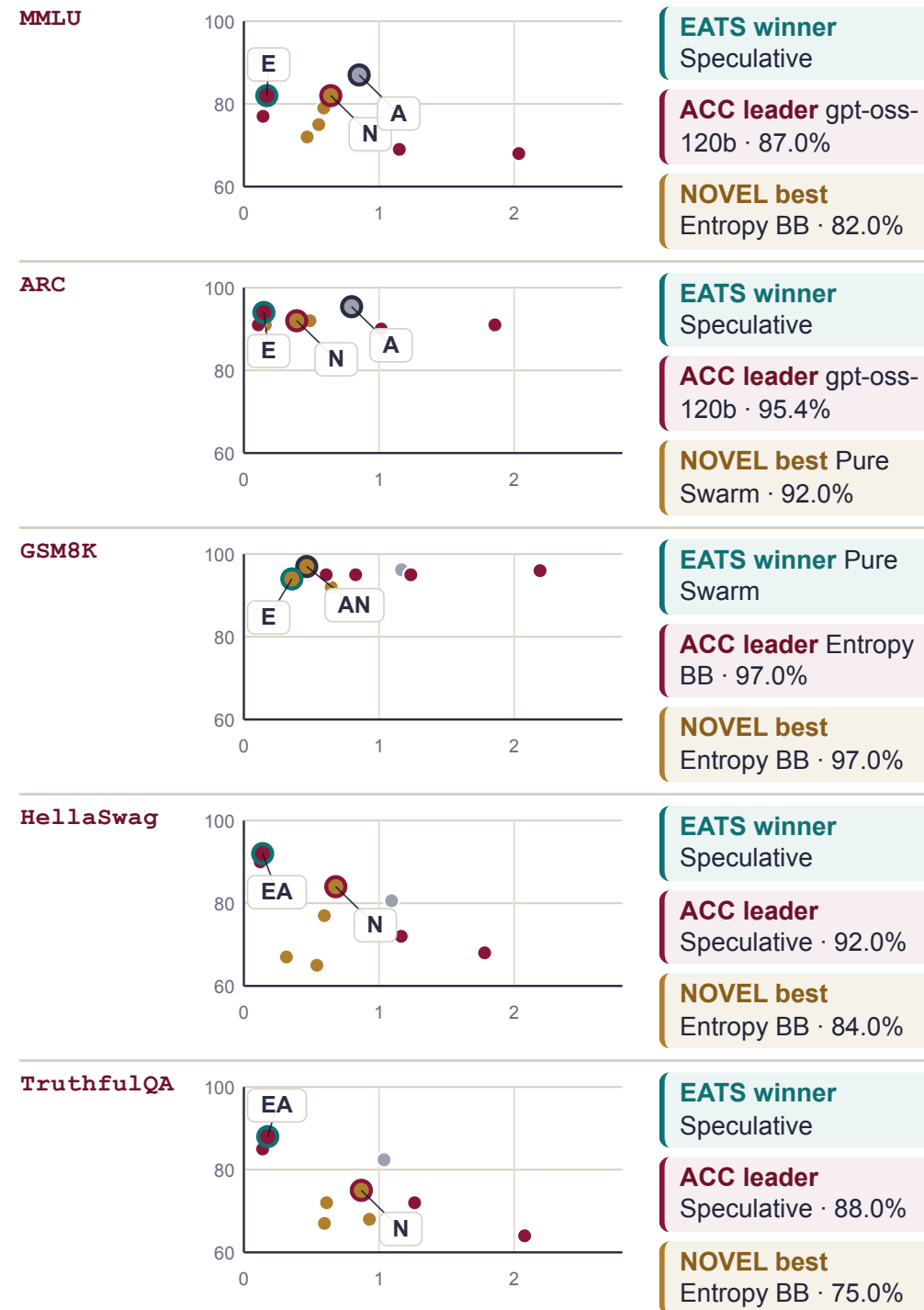
Ten architectures in three families. † marks novel project contributions introduced in the current study.

FAMILY	ARCHITECTURE	KEY MECHANISM
STAND	Single-Model	direct inference
STAND	MoE	internal sparse expert activation
HYBRID	Routing	confidence SLM→LLM escalation
HYBRID	Speculative	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	proponent–opponent–arbitrator
HYBRID	Active Oracle †	targeted LLM sub-queries
HYBRID	Blackboard †	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	uncertainty-penalised bidding
MULTI	Ensemble	majority or confidence-weighted voting
MULTI	Pure Swarm †	SLM-only peer coordination; no LLM online



8 Benchmark Readouts

Each mini-plot shows accuracy against efficiency penalty, so the strongest trade-offs appear toward the upper-left. **E** = EATS winner, **A** = accuracy leader, **N** = best novel result.



9 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	0.80	0.90	0.78	0.89	0.85
Speculative	0.82	0.91	0.72	0.90	0.86
Standalone (MoE)	0.68	0.73	0.66	0.59	0.61
Blackboard †	0.69	0.80	0.82	0.67	0.55
Entropy BB †	0.69	0.79	0.83	0.69	0.60
Active O. †	0.67	0.89	0.75	0.67	0.64
Debate POA	0.52	0.66	0.64	0.53	0.52
Pure Swarm †	0.67	0.82	0.84	0.60	0.61
Ensemble	0.40	0.53	0.52	0.43	0.38

Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

10 Conclusions

RQ1 Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.

RQ2 Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.

RQ3 EATS gives a consistent cross-architecture ranking: speculative leads four benchmarks, and Pure Swarm leads GSM8K.

RQ4 The measurements corroborate 8 of 10 SLR trends: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

△ LIMITATIONS

- Single hardware stack (L4 / PRO6000 / H200) — results may not generalise to other GPU tiers or cloud inference endpoints.
- Benchmark scope covers 5 NLP task types; code generation, retrieval-augmented and long-context tasks are not included.
- EATS weights ($\beta = 0.60$, $\lambda = 0.40$) are fixed after calibration; sensitivity to alternative weight choices is not explored in this study.

SELECTED REFERENCES

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Companion SLR (54 studies, PRISMA 2020) · [6] MMLU, ARC, GSM8K, HellaSwag, TruthfulQA.

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